Performance Testing

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the system behaves under expected and peak load conditions. The prediction endpoint is now session-authenticated, so every test run below signs in first (via a Postman/curl login request, reusing the returned session cookie) before exercising /predict.

Step 1: Testing Overview

Testing Tool Used	Postman, curl (manual, with session-cookie login); can be extended with Locust/k6 for load testing
Type of Testing	Functional + manual load spot-checks
Target Module / API	POST /login, then POST /predict (login_required)
Test Environment	Local (Flask dev server) and Render (Gunicorn, cloud)
Test Date	03 July 2026

Step 2: Test Scenarios

S.NO	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Attempt POST /predict while logged out	1	<1	302 redirect to /login (not a 200/crash)
2	Log in, then submit valid high-rainfall values (flood-likely case)	1	<1	Returns chance.html with high-risk gauge; row saved to /history
3	Log in, then submit valid low-rainfall values (no-flood case)	1	<1	Returns no_chance.html with low-risk gauge; row saved to /history
4	Log in, submit form with a missing required field	1	<1	WTForms validation error shown inline, no server crash
5	Submit POST /predict with a missing/invalid CSRF token	1	<1	400 Bad Request (CSRF check fails safely)
6	Repeated sequential authenticated submissions (basic load spot-check)	10	10	All requests handled without errors or slowdown

Step 3: Performance Test Results

S. N O	Metric	Target Value	Actual Value	Status	Remarks
1	**Response Time (Avg, /predict)**	< 2 seconds	< 0.3 seconds	Pass	Model inference is lightweight (classical ML, no GPU needed)
2	**Response Time (Max, /predict)**	< 5 seconds	< 0.5 seconds	Pass	Observed on local dev server
3	**Login / session overhead**	< 0.5 seconds	< 0.1 seconds	Pass	Werkzeug password hashing adds negligible latency at this scale
4	**Throughput (Req/sec)**	-	Not formally load-tested	N/A	Recommend Locust/k6 pass before production traffic
5	**Error Rate**	< 1%	0% (valid, authenticated inputs)	Pass	Only missing-field or missing-CSRF requests intentionally error
6	**CPU Utilization**	< 80%	Low	Pass	Classical ML inference is CPU-cheap
7	**Memory Utilization**	< 80%	Low	Pass	Model + scaler files are small (<60 KB combined); SQLite adds minimal overhead

Step 4: Observations & Analysis

The app performs well under light, manual testing since the model is a classical scikit-learn classifier rather than a deep learning model, keeping inference time negligible. Adding authentication introduced one new consideration: every automated test now needs a valid session (or a fresh CSRF token) before it can reach /predict, so test scripts log in first and reuse the returned cookie for subsequent requests. Formal load testing (concurrent virtual users via Locust or k6, including concurrent authenticated sessions) was not yet performed and is recommended before scaling to production traffic.

Step 5: Screenshots / Evidence

```
OUTPUT   TERMINAL   PORTS   AZURE                                   powershell
(retrain_env) PS C:\Users\lahar\OneDrive\Desktop\Rising-Waters> Get-Content register_result.html -Raw


        <form action="/register" method="POST" novalidate>
            <input id="csrf_token" name="csrf_token" type="hidden" value="IjgwNjQwMTQyYmUxODQ2MjMyODNkZDc0ZjkxY2M0OTUwMTE4ODM1YmUi.alY9qw.JAWxjwdCb8mypI4RvpuwgG1CgvI
">

            <div class="field-grid" style="grid-template-columns:1fr;">
                <div class="field">
                    <label for="username">Username</label>
                    <input autocomplete="username" id="username" maxlength="64" minlength="3" name="username" placeholder="Pick a username" required type="text" valu
e="testuser">
                    <span class="field-error">That username is already taken.</span>
                </div>
                <div class="field">
                    <label for="email">Email</label>
                    <input autocomplete="email" id="email" maxlength="120" name="email" placeholder="you@example.com" required type="text" value="testuser@example.co
m">
                    <span class="field-error">An account with that email already exists.</span>
                </div>
                <div class="field">
                    <label for="password">Password</label>
                    <input autocomplete="new-password" id="password" minlength="8" name="password" placeholder="At least 8 characters" required type="password" value
="">

                </div>
                <div class="field">
                    <label for="confirm_password">Confirm password</label>
                    <input autocomplete="new-password" id="confirm_password" name="confirm_password" placeholder="Repeat your password" required type="password" valu
e="">

                </div>
            </div>

            <div class="form-footer">
                <span class="hint">Already have an account? <a href="/login">Log in</a></span>
                <input class="btn" id="submit" name="submit" type="submit" value="Create account">
            </div>
        </form>
    </div>
```

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OUTPUT TERMINAL PORTS AZURE powershell

```
(retrain_env) PS C:\Users\lahar\OneDrive\Desktop\Rising-Waters> Get-Content home.html -Raw
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Flood Prediction System</title>
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link href="https://fonts.googleapis.com/css2?family=Space+Grotesk:wght@500;700&family=IBM+Plex+Sans:wght@400;500&family=IBM+Plex+Mono:wght@400;500&display=swap"
 rel="stylesheet">
    <link rel="stylesheet" href="/static/main.css">
    <style>
        /* --- Extra weather effect layer: wind + ripples (self-contained) --- */
        #weather-fx-canvas {
            position: fixed;
            top: 0;
            left: 0;
            width: 100vw;
            height: 100vh;
            pointer-events: none;
            z-index: 2; /* above background clouds, below panel content */
        }
    </style>
</head>
<body data-rain="normal">

<canvas id="rain-canvas"></canvas>
<canvas id="weather-fx-canvas"></canvas>
<div class="cloud-layer">
    <div class="cloud c1"></div>
    <div class="cloud c2"></div>
    <div class="cloud c3"></div>
</div>
<div class="lightning-flash"></div>

<nav class="navbar">
    <div class="nav-brand"><span class="dot"></span>Floods Prediction</div>
    <div class="nav-links">
        <a href="/" class="active">Home</a>
        <a href="/predict" class="">Predict Floods</a>
```

OUTPUT TERMINAL PORTS AZURE powershell

```
(retrain_env) PS C:\Users\lahar\OneDrive\Desktop\Rising-Waters> Get-Content predict_result.html -Raw
    </div>

    <div class="panel">
        <span class="status-icon">âš ï¸</span>
        <span class="eyebrow" style="color:var(--amber);">Assessment Complete</span>
        <h1 class="title-hero" style="font-size:32px;">Flood Predicted</h1>

        <div class="gauge-wrap" data-gauge data-risk="high" data-value="86">
            <svg class="gauge" viewBox="0 0 260 150">
                <path class="gauge-track" d="M 10 130 A 120 120 0 0 1 250 130" />
                <path class="gauge-fill" d="M 10 130 A 120 120 0 0 1 250 130" />
                <line class="gauge-needle" x1="130" y1="130" x2="130" y2="30" stroke="#E7F1F8" stroke-width="3" stroke-linecap="round" />
                <circle class="gauge-center" cx="130" cy="130" r="6" />
            </svg>
            <div class="gauge-readout status-flood">
                Risk level
                <strong>High</strong>
            </div>
        </div>

        <p>
            The rainfall and weather values you entered fall into a range
            the model associates with a high likelihood of flooding.
            Treat this as an early indicator, not a certainty â€” check local
            advisories for confirmed conditions.
        </p>

        <a href="/predict">
            <button class="btn" type="button">Predict Again</button>
        </a>

        <div class="foot-note">Estimate based on entered parameters only</div>
    </div>

</div>

<script src="/static/main.js"></script>
<footer class="site-footer">
    <span>&copy; 2026 Flood Prediction System</span>
    <span class="footer-dot">&middot;</span>
```

OUTPUT TERMINAL PORTS AZURE powershell

```
(retrain_env) PS C:\Users\lahar\OneDrive\Desktop\Rising-Waters> Get-Content predict_error.html -Raw
<div class="page">

    <div class="panel wide">
        <span class="eyebrow">Enter Readings</span>
        <h2 class="title-hero" style="font-size:28px;">Weather &amp; Rainfall Details</h2>
        <p style="margin-bottom:0;">Fill in each field with the local values for the period you want to assess.</p>


        <form action="/predict" method="POST" novalidate>
            <input id="csrf_token" name="csrf_token" type="hidden" value="ImFhOGM0MDYwNTg2MTQxY2ExNzk3NjU2NmQ1Mzg2ZDZiODgyOGQ1NDAi.alZNUQ.OrWlY5y2Ps5TqkbsgTRwx6-TBek
">

            <div class="section-label">Weather conditions</div>
            <div class="field-grid">
                <div class="field">
                    <label for="Temp">Temperature (Â°C)</label>
                    <input id="Temp" name="Temp" placeholder="Average Temperature" required step="any" type="text" value="">
                    <span class="field-error">This field is required.</span>
                </div>
                <div class="field">
                    <label for="Humidity">Humidity (%)</label>
                    <input id="Humidity" name="Humidity" placeholder="Humidity Percentage (%)" required step="any" type="text" value="74">

                </div>
                <div class="field">
                    <label for="CloudCover">Cloud Cover (%)</label>
                    <input id="CloudCover" name="CloudCover" placeholder="Cloud Cover Percentage (%)" required step="any" type="text" value="40">

                </div>
            </div>

            <div class="section-label">Rainfall totals</div>
            <div class="field-grid">
                <div class="field">
                    <label for="Annual">Annual Rain Fall (mm)</label>
                    <input id="Annual" name="Annual" placeholder="Enter Annual Rain Fall in mm" required step="any" type="text" value="3671.1">

                </div>
                <div class="field">
                    <label for="JanFeb">Jan-Feb Rainfall (mm)</label>
```

OUTPUT TERMINAL PORTS AZURE powershell

```
(retrain_env) PS C:\Users\lahar\OneDrive\Desktop\Rising-Waters> Get-Content login_bad_result.html -Raw

<nav class="navbar">
    <div class="nav-brand"><span class="dot"></span>Floods Prediction</div>
    <div class="nav-links">
        <a href="/" class="">Home</a>
        <a href="/predict" class="">Predict Floods</a>

            <a href="/login" class="active">Log in</a>
            <a href="/register" class="">Sign up</a>

        <button class="theme-toggle" id="theme-toggle" type="button" aria-label="Toggle light and dark theme">
            <span class="theme-toggle-icon">ðŸŒ™</span>
        </button>
    </div>
</nav>


        <div class="flash-stack">

                <div class="alert alert-error">Incorrect email or password.</div>

        </div>



<div class="page">

    <div class="panel">
        <span class="eyebrow">Welcome back</span>
        <h2 class="title-hero" style="font-size:28px;">Log In</h2>



        <form action="/login" method="POST" novalidate>
            <input id="csrf_token" name="csrf_token" type="hidden" value="ImUwZmY3NDJkNjdiYWE0ODc0YjY0YzZiMzk1ZGMxOGJkMDkwNGRkYzci.alZONg.J5p898sQeuD7l12yC1QnZqVh_As
">

            <div class="field-grid" style="grid-template-columns:1fr;">
                <div class="field">
                    <label for="email">Email</label>
```

OUTPUT TERMINAL PORTS AZURE powershell

```
(retrain_env) PS C:\Users\lahar\OneDrive\Desktop\Rising-Waters> Get-Content predict_noauth.html -Raw

<div class="page">

    <div class="panel">
        <span class="eyebrow">Welcome back</span>
        <h2 class="title-hero" style="font-size:28px;">Log In</h2>



        <form action="/login" method="POST" novalidate>
            <input id="csrf_token" name="csrf_token" type="hidden" value="IjMwZmRmODQ3ZTEzZTQyNGEyNWIzZDEyYmVmYzdlNzRkMjBhOTE5MmEi.alZOYg.Br4bQ3rbNgKIq23pCa0H86LdyW4
">

            <div class="field-grid" style="grid-template-columns:1fr;">
                <div class="field">
                    <label for="email">Email</label>
                    <input autocomplete="email" id="email" name="email" placeholder="you@example.com" required type="text" value="">

                </div>
                <div class="field">
                    <label for="password">Password</label>
                    <input autocomplete="current-password" id="password" name="password" placeholder="Your password" required type="password" value="">

                </div>
                <div class="field field-checkbox">
                    <input id="remember" name="remember" type="checkbox" value="y"> <label for="remember">Keep me signed in</label>
                </div>
            </div>

            <div class="form-footer">
                <span class="hint">New here? <a href="/register">Create an account</a></span>
                <input class="btn" id="submit" name="submit" type="submit" value="Log in">
            </div>
        </form>
    </div>

</div>

<script src="/static/main.js"></script>
<footer class="site-footer">
```

OUTPUT TERMINAL PORTS AZURE powershell

```
(retrain_env) PS C:\Users\lahar\OneDrive\Desktop\Rising-Waters> Get-Content logout_result.html -Raw
    // Spawn a new ripple every ~250ms
    setInterval(spawnRipple, 250);

    function drawRipples() {
        for (let i = ripples.length - 1; i >= 0; i--) {
            const r = ripples[i];
            ctx.beginPath();
            ctx.strokeStyle = `rgba(150, 200, 220, ${r.opacity})`;
            ctx.lineWidth = 1.2;
            ctx.ellipse(r.x, r.y, r.radius, r.radius * 0.35, 0, 0, Math.PI * 2);
            ctx.stroke();

            r.radius += 0.6;
            r.opacity -= 0.012;

            if (r.radius >= r.maxRadius || r.opacity <= 0) {
                ripples.splice(i, 1);
            }
        }
    }

    function loop() {
        ctx.clearRect(0, 0, canvas.width, canvas.height);
        drawWind();
        drawRipples();
        requestAnimationFrame(loop);
    }
    loop();
})();
</script>
<footer class="site-footer">
    <span>&copy; 2026 Flood Prediction System</span>
    <span class="footer-dot">&middot;</span>
    <span>Built by Lahari Sri</span>
    <span class="footer-dot">&middot;</span>
    <span>Made with Flask &amp; scikit-learn</span>
</footer>
</body>
</html>
```